

Mathematical and Computational Methods

Exercise Sheet 4: Vectors

This sheet accompanies *Unit 4 (Vectors)*. All methods are those of Unit 4 only—vector algebra, the dot and cross products, and the equations of lines and planes; nothing from later units is needed.

How to use this sheet. Problems are graded by difficulty—★ drill (routine fluency), ★★ core (standard, tutorial level) and ★★ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ►) and **further practice** for independent home study.

Solutions. A full worked solution sits after each problem: hidden in the student build, shown in the solutions build. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showolutions{}\input{MCM_Exercises_Unit4}"`) and recompile.

Notation. Vectors are bold, $\mathbf{u}, \mathbf{v}, \mathbf{w}$; $\|\mathbf{v}\|$ is the magnitude, $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$ a unit vector, $\mathbf{u} \cdot \mathbf{v}$ the dot product, $\mathbf{u} \times \mathbf{v}$ the cross product, and $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}$ the projection of $\mathbf{u}$ onto $\mathbf{v}$. The standard basis of $\mathbb{R}^3$ is $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

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1 Warm-up drills

Short, routine problems — at least one per skill — to build fluency. Answers should be quick.

Exercise 1.1 (*★ drill — component arithmetic*). Let $\mathbf{u} = (3, -1, 2)$ and $\mathbf{v} = (-1, 4, 0)$. Compute

$$(a) \mathbf{u} + \mathbf{v}, \quad (b) \mathbf{u} - \mathbf{v}, \quad (c) 3\mathbf{u}, \quad (d) 2\mathbf{u} - 3\mathbf{v}.$$

Exercise 1.2 (*★ drill — vector between points, distance*). For the points $A(2, -1, 3)$ and $B(5, 1, -1)$, find (a) the vector $\overrightarrow{AB}$ and (b) the distance from A to B .

Exercise 1.3 (*★ drill — unit vectors*). Normalise each vector: (a) $\mathbf{v} = (3, 4)$, (b) $\mathbf{w} = (1, -2, 2)$.

Exercise 1.4 (*★ drill — dot product and orthogonality*). Let $\mathbf{a} = (2, 3)$, $\mathbf{b} = (3, -2)$, $\mathbf{c} = (1, 1)$. Compute $\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{c}$ and $\mathbf{b} \cdot \mathbf{c}$, and say which pair (if any) is orthogonal.

Exercise 1.5 (*★ drill — angle via the dot product*). Find the angle between (a) $(3, 0)$ and $(0, -5)$; (b) $(1, 1)$ and $(-1, 0)$.

Exercise 1.6 (*★ drill — projection*). Project $\mathbf{u} = (6, 2)$ (a) onto $\mathbf{v} = (3, 0)$ and (b) onto $\mathbf{w} = (0, 5)$.

Exercise 1.7 (*★ drill — cross product*). Compute (a) $\mathbf{i} \times \mathbf{j}$ and (b) $(1, 2, 3) \times (4, 5, 6)$.

Exercise 1.8 (*★ drill — area*). Find the area of the parallelogram spanned by $\mathbf{u} = (2, -2, 1)$ and $\mathbf{v} = (1, 2, 2)$, and hence the area of the triangle with these as two sides.

Exercise 1.9 (*★ drill — equation of a line*). (a) Give the vector, parametric and symmetric equations of the line through $P(1, 2, 3)$ in the direction $\mathbf{d} = (2, -1, 4)$. (b) Give the vector equation and the slope of the line through $A(0, 1)$ and $B(4, 3)$ in $\mathbb{R}^2$.

Exercise 1.10 (*★ drill — equation of a plane*). (a) Find the Cartesian equation of the plane through $(2, -1, 3)$ with normal $\mathbf{n} = (1, 2, -1)$. (b) Write down a normal vector to the plane $3x - y + 2z = 7$.

Exercise 1.11 (*★ drill, conceptual — true or false*). State whether each is true or false, with a one-line reason.

1. If $\mathbf{u} \cdot \mathbf{v} = 0$ then $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$.
2. $\mathbf{u} \times \mathbf{v} = \mathbf{v} \times \mathbf{u}$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$.
3. $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ for every $\mathbf{v} \in \mathbb{R}^3$.
4. The dot product of two vectors is a vector.
5. $\|\mathbf{u} + \mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$ for all $\mathbf{u}, \mathbf{v}$.

Exercise 1.12 (*★ drill — sketching sums and differences*). Take $\mathbf{u} = (3, 1)$ and $\mathbf{v} = (1, 2)$ in $\mathbb{R}^2$. On a single set of axes, sketch $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$, each drawn from the origin. Read the two combinations off your sketch and confirm them by component arithmetic.

2 Tutorial problems

The core set for the 2-hour tutorial: standard difficulty, anchored on the unit's worked examples, with problems aimed at each cohort (mathematics, data science, physics). Problems flagged ► form the live *tutorial core* to be worked in the 2-hour class; the remaining tutorial problems are for completion at home.

Exercise 2.1 (► ★★core — *linear combinations*). Let $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (3, -1)$.

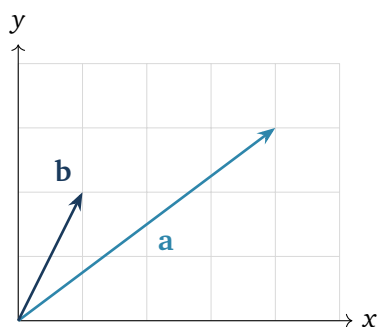
1. Express $\mathbf{c} = (7, 0)$ as $\alpha\mathbf{a} + \beta\mathbf{b}$.
2. Are $\mathbf{a}$ and $\mathbf{b}$ linearly independent? Explain geometrically.

Exercise 2.2 (► ★★core, data science — *angle and cosine similarity*). Over the four-word vocabulary (data, model, train, plot) two short documents have word-count vectors $\mathbf{u} = (1, 1, 0, 0)$ and $\mathbf{v} = (0, 1, 1, 0)$.

1. Find the angle θ between $\mathbf{u}$ and $\mathbf{v}$ using the dot product.
2. State their cosine similarity $\cos \theta$ and interpret it: would these documents be judged similar?

Exercise 2.3 (► ★★core — *parallel and perpendicular components*). Resolve $\mathbf{u} = (2, 4)$ into a component parallel to $\mathbf{v} = (1, 1)$ and a component perpendicular to it. Verify the two pieces are orthogonal and sum to $\mathbf{u}$.

Exercise 2.4 (► ★★core — *reading a diagram*). The figure shows two vectors $\mathbf{a}$ and $\mathbf{b}$ drawn from the origin on a unit grid.



1. Read off the components of $\mathbf{a}$ and $\mathbf{b}$ from the grid.
2. Compute the projection $\text{proj}_{\mathbf{b}} \mathbf{a}$ of $\mathbf{a}$ onto $\mathbf{b}$.

Exercise 2.5 (► ★★core — *a perpendicular direction*). Find a vector orthogonal to both $\mathbf{a} = (1, 2, 1)$ and $\mathbf{b} = (2, -1, 3)$, and then a *unit* vector in that direction.

Exercise 2.6 (★★core — *area of a triangle in space*). Find the area of the triangle with vertices $A(1, 1, 0)$, $B(2, 3, 1)$ and $C(0, 2, 2)$.

Exercise 2.7 (► ★★core — *a line, and points on it*). Let ℓ be the line through $A(1, 0, 2)$ and $B(3, 4, 6)$.

1. Give a parametric equation of ℓ , using the simplest integer direction.
2. Does $C(0, -2, 0)$ lie on ℓ ?
3. Does $D(2, 1, 4)$ lie on ℓ ?

Exercise 2.8 (► ★★core — *the plane through three points (anchor)*). Find the equation of the plane through $A(2, 0, 0)$, $B(0, 3, 0)$ and $C(0, 0, 6)$, in both point-normal and Cartesian form, and verify all three points satisfy it.

Exercise 2.9 (★★core — *the same plane in parametric form*). Using the same three points $A(2, 0, 0)$, $B(0, 3, 0)$, $C(0, 0, 6)$ as above:

1. Write the plane in parametric (vector) form $\mathbf{r} = A + s\overrightarrow{AB} + t\overrightarrow{AC}$.
2. Evaluate the point at $s = 1$, $t = 1$ and verify it satisfies the Cartesian equation $3x + 2y + z = 6$ found above.

Exercise 2.10 (★★core — *line meets plane*). Where does the line $\mathbf{r} = t(1, 1, 1)$ meet the plane $3x + 2y + z = 6$ found above?

Exercise 2.11 (► ★★core, physics — *torque and work*). A force $\mathbf{F} = (2, 3, 1)$ (newtons) acts at the point with position vector $\mathbf{r} = (1, 0, 2)$ (metres) measured from a pivot.

1. Find the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ about the pivot.
2. The point of application then undergoes the displacement $\mathbf{d} = (0, 4, 0)$ (metres). Find the work $W = \mathbf{F} \cdot \mathbf{d}$ done by the force.

Exercise 2.12 (★★core, data science — *ranking by cosine similarity*). A query is the vector $\mathbf{q} = (1, 1, 1)$. Three documents are $\mathbf{d}_1 = (2, 2, 2)$, $\mathbf{d}_2 = (1, 1, 0)$, $\mathbf{d}_3 = (0, 0, 5)$. Rank them by cosine similarity to $\mathbf{q}$, and comment on $\mathbf{d}_1$.

Exercise 2.13 (★★core, enrichment — *volume and coplanarity*). 1. Find the volume of the parallelepiped on $\mathbf{u} = (1, 0, 1)$, $\mathbf{v} = (2, 3, 0)$, $\mathbf{w} = (1, 1, 1)$ using the scalar triple product $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$.

2. Show that $\mathbf{a} = (1, 1, 1)$, $\mathbf{b} = (2, 1, 0)$, $\mathbf{c} = (0, 1, 2)$ are coplanar.

Exercise 2.14 (★★core, conceptual — *spot the error*). A student writes: “For $\mathbf{u} = (1, 2, 2)$ and $\mathbf{v} = (2, 2, 1)$ we have $\mathbf{u} \cdot \mathbf{v} = 2 + 4 + 2 = 8$, so $\cos \theta = 8$ and $\theta = \arccos 8$.” Identify the mistake and give the correct angle.

3 Further practice (home)

A larger bank for independent study, from routine to harder, ending with a few **challenge** problems (★ ★ ★). Levels are tagged on each.

Exercise 3.1 (★ drill — *parallel vectors*). Find the scalar k for which $(k, 6)$ is parallel to $(2, 3)$.

Exercise 3.2 (★ drill — *magnitude in higher dimensions*). For $\mathbf{v} = (1, 1, 1, 1) \in \mathbb{R}^4$ find $\|\mathbf{v}\|$ and the unit vector $\hat{\mathbf{v}}$.

Exercise 3.3 (★ drill — *perimeter from distances*). Find the perimeter of the triangle with vertices $A(0, 0)$, $B(3, 0)$, $C(0, 4)$.

Exercise 3.4 (★★ core — *unit vectors at a given angle*). Find all unit vectors in $\mathbb{R}^2$ that make an angle of 45° with $(1, 0)$.

Exercise 3.5 (★★ core — *lengths from dot products*). Given $\|\mathbf{u}\| = 3$, $\|\mathbf{v}\| = 5$ and $\mathbf{u} \cdot \mathbf{v} = 6$, find $\|\mathbf{u} + \mathbf{v}\|$ and $\|\mathbf{u} - \mathbf{v}\|$.

Exercise 3.6 (★★ core — *angle in a triangle*). For the triangle $A(1, 0)$, $B(3, 2)$, $C(0, 1)$, find the interior angle at vertex A .

Exercise 3.7 (★★ core — *solve for orthogonality*). Find k so that $(k, 2, 1)$ and $(3, k, -5)$ are orthogonal.

Exercise 3.8 (★★ core — *resolve into components*). Resolve $\mathbf{u} = (3, 4)$ into parts parallel and perpendicular to $\mathbf{v} = (2, 1)$.

Exercise 3.9 (★★ core — *anticommutativity and area*). For $\mathbf{u} = (2, 1, -1)$, $\mathbf{v} = (1, 3, 2)$, compute $\mathbf{u} \times \mathbf{v}$ and $\mathbf{v} \times \mathbf{u}$, confirm they are negatives, and give the area of the triangle on $\mathbf{u}, \mathbf{v}$.

Exercise 3.10 (★★ core — *unit normal to a plane*). Find a unit normal to the plane through $P(1, 0, 0)$, $Q(0, 1, 0)$, $R(0, 0, 1)$, and the Cartesian equation of the plane.

Exercise 3.11 (★★ core — *intersection of two lines*). Do the lines $\ell_1 : \mathbf{r} = t(1, 2)$ and $\ell_2 : \mathbf{r} = (3, 0) + s(-1, 1)$ meet? If so, where?

Exercise 3.12 (★★ core — *parallel planes*). Are the planes $2x - y + 3z = 5$ and $-4x + 2y - 6z = 1$ parallel? Do they intersect?

Exercise 3.13 (★★ core — *plane parallel to a given one*). Find the plane through $(1, 2, 3)$ parallel to $2x - y + 2z = 0$.

Exercise 3.14 (★★ core — *angle between two planes*). Find the acute angle between the planes $x + y + z = 1$ and $x + z = 0$.

Exercise 3.15 (★★ core — *distance from a point to a plane*). Find the distance from the point $P(3, 1, 2)$ to the plane $2x - 2y + z = 4$, and the foot F of the perpendicular from P to the plane.

Exercise 3.16 (★★ core, physics — *equilibrium of forces*). Two forces $\mathbf{F}_1 = (2, 3)$ and $\mathbf{F}_2 = (-1, 4)$ (newtons) act on a particle. Find the third force $\mathbf{F}_3$ that holds it in equilibrium, and its magnitude.

Exercise 3.17 (★★ core, data science — *orthogonal preferences*). Two users rate four films, giving rating-deviation vectors $\mathbf{a} = (2, 2, 2, 2)$ and $\mathbf{b} = (1, -1, 1, -1)$. Compute their cosine similarity and interpret it.

Exercise 3.18 (★★core, physics — *work done sliding down a ramp*). A crate of mass 10 kg slides down a ramp, undergoing the displacement $\mathbf{s} = (6, -8)$ (metres), with the $+y$ axis pointing up. Taking $g = 10 \text{ m/s}^2$, the weight is the force $\mathbf{F} = (0, -100)$ (newtons).

1. Find the work $W = \mathbf{F} \cdot \mathbf{s}$ done by gravity.
2. Find the component of the weight in the direction of motion.

Exercise 3.19 (★★core — *a “show that” bridge to the proofs*). These identities power the two proofs that follow; establish them first. Using only $\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w}$, the distributive rule $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$, the scalar rule $(\lambda \mathbf{u}) \cdot \mathbf{v} = \lambda(\mathbf{u} \cdot \mathbf{v})$ and commutativity $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$, show that for all $\mathbf{u}, \mathbf{v}$

$$(a) \quad \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2, \quad (b) \quad \|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2.$$

Exercise 3.20 (★★★challenge — *the parallelogram law*). Using only the properties of the dot product, prove that for all $\mathbf{u}, \mathbf{v}$

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2.$$

Exercise 3.21 (★★★challenge — *diagonals of a rhombus*). A rhombus has sides $\mathbf{u}$ and $\mathbf{v}$ with $\|\mathbf{u}\| = \|\mathbf{v}\|$. Its diagonals are $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$. Prove the diagonals are perpendicular.

Exercise 3.22 (★★★challenge — *coplanarity of four points*). Decide whether the points $P(0, 0, 0)$, $Q(1, 1, 0)$, $R(1, 0, 1)$, $S(0, 1, 1)$ are coplanar. If not, give the volume of the parallelepiped on $\vec{PQ}, \vec{PR}, \vec{PS}$.

Exercise 3.23 (★★★challenge — *foot of the perpendicular*). Let ℓ be the line through the origin in the direction $\mathbf{d} = (1, 2)$, and let $P(5, 0)$. Find the foot F of the perpendicular from P to ℓ , and the distance from P to ℓ .